

Harvesting and Operations

Assessing the Economic Viability of the Mechanized Removal of Understory Beech during a Shelterwood Harvest

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Abstract

Beech bark disease is a pathogenic complex that has been spreading throughout the American beech's range since the 1800s. A litany of negative consequences have manifested from the infestation of this disease, many of which deteriorate the ecological functions of forestland. This case study sought to analyze the cost structure for removing a recalcitrant beech understory via mechanized shelterwood harvesting. High-resolution data regarding the day-to-day operation of harvesting equipment was collected using daily production journals. Interviews were conducted with the logging company owner and maintenance supervisor to gather additional information required to calculate machine costs, overhead, job specific costs, and trucking costs. The yield from this harvest was 527 metric tonnes of sawtimber and 4,893 tonnes of clean chips. The total harvesting cost equated to \$4,651/ha, with the cost attributed to removing beech at \$204/ha. Despite the additional cost of beech removal, the logger generated a total profit of \$5,965 and a return on investment of 7.5%, allowing us to conclude that mechanized harvesting can be a viable beech removal strategy given the forest stocking and market conditions that are in place.

Study Implications: This study breaks down the various costs associated with cutting, skidding, landing, and transporting wood products from a mechanized harvesting operation designed to remediate the effects of beech bark disease. The beech remediation harvest was economically viable for both the landowner and the logger because the timber sale included some valuable hardwood sawtimber, and the harvest system was capable of generating clean chips for a pulp mill with the low-grade hardwood. Furthermore, the landowner's willingness to accept lower sawtimber stumpage revenues allowed the logger to make a profit and return on investment on the job. Had the timber sale been limited to only hardwood pulpwood or fuel chips, the operation would not have been economically viable without the landowner paying for the operation, which, based on our analysis, would be approximately \$200/ha.

Keywords: beech bark disease, shelterwood, cost structure, return on investment

Beech bark disease (BBD) is a pathogenic insect-fungus complex spread by the feeding of bark scale insects (*Cryptococcus fagisugae* Lind. and *Xylococcus betulae* Perg.) and the subsequent infection of *Neonectria faginata* and *N. ditissima* (Ehrlich 1934, Houston et al. 1979, Houston and O'Brien 1983, Cale et al. 2017). Arriving in Nova Scotia before 1900 (Hewitt 1914, Hawboldt 1944), BBD has since spread through much of the American beech's (beech) (*Fagus grandifolia* Ehrh.) range, reaching the Great Smoky Mountains National Park by 1986 and confirmed in Michigan by 2000 (Witter et al. 2004, Rumble et al. 2020). Shigo (1972) identified three phases of BBD: an advance front, in which most trees become infested with beech scale to some degree; a killing front, when widespread infection-induced mortality occurs; and aftermath forests, comprised of dense beech thickets and severely deformed trees of low vigor. Thickets are believed to develop from a combination of root sprouts and seedlings that proliferate after a decline in the beech canopy due to reduced vitality (Houston 1975, Giencke et al. 2014). Root sprouts exhibit up to triple the lateral extension of seedlings (Beaudet and Messier 2008)

and thickets create light conditions too shaded to successfully regenerate sugar maple (*Acer saccharum* Marsh.) (Hane 2003), one of the most shade-tolerant and important species associated with beech. Reductions in native biodiversity, species richness, and aboveground biomass have been recorded in BBD aftermath forests and are the result of altered forest structure and reduced vigor of mature trees (Forrester et al. 2003, Cale et al. 2013, 2015, Rumble et al. 2020).

Without intervention, BBD infestations can persist in beech trees intergenerationally for decades (Houston 1975, Cale and McNulty 2018). Under the premise of climate change, beech susceptibility to infection as well as scale insect and fungal survivorship are expected to increase (Stephanson and Coe 2017, Jactel et al. 2019). Given the prospect of increasingly stressful conditions for beech and increasing spread of BBD and its current extent of infection, a failure to take managerial action will ensure the loss of ecosystem functions for generations. Beech is an important component in the most abundant forest cover type of the North, maple-beech-birch, doubling in size to over 20 million ha from 1953 and 1997,

and projected to continue growing at a similar rate until 2050 (Alig and Butler 2004). Forest remediation will be necessary, as this issue is only expected to spread, with no evidence to suggest it will dissipate naturally.

Despite its century-long presence in North America, little has been done to reverse the impacts of BBD. Although the biology and ecology of BBD is well understood, significant reviews of which are provided by Houston (1994) and Cale et al. (2017), any breakthroughs in management strategies have been unsuccessful in instigating widespread implementation. Herbicides are effective in removing beech thickets (Ostrofsky and McCormack 1986), but treating forests on a landscape scale can be prohibitively expensive for even large landowners and, given the negative perception of herbicides by the public (Wagner et al. 1998), coordinated efforts by smaller landowners is unlikely. Brush-saws are effective for removing beech thickets but would similarly involve exorbitantly high costs because of the broad expanse of thickets (Forrester and Bohn 2007). Several studies have investigated the efficacy of harvesting methods to reduce the presence of BBD and dense understories (Filip 1978, Mielke et al. 1986, Houston 2001, Bohn and Nyland 2003, Nyland 2004, Farrar and Ostrofsky 2006, Leak 2006), yet none have explicitly examined the financial implications of mechanical removal. It is possible that the cost will require harvesting to be an investment by a landowner in their forest property, but as Ostrofsky (2004) pointed out, the benefits of reclamation to future productivity are likely to outweigh the initial costs after the first rotation. Completing a harvest that does not provide sufficient revenue to cover costs is not economically viable for logging firms, regardless of stumpage or contract rates. Even if a landowner were to contract a logger to implement a silvicultural prescription designed to remove recalcitrant understory beech, the logger may still abstain because the costs could be too high to be covered by payments from landowners or mills (Germain et al. 2016, Regula et al. 2018, Germain et al. 2019).

Some of the earliest studies of mitigating the effects BBD took place at the Bartlett Experimental Forest in New Hampshire. In a 1978 Forest Service report, Filip (1978) presented findings from two successive, single-tree selection harvests that occurred in 1952 and 1975. There was no cost analysis involved, but the report indicated that the harvests successfully increased the proportion of merchantable timber in the stand as well as improved the proportion of grade 1 and 2 logs among the nonbeech sawtimber. There is no indication that single-tree selection is effective in reducing the presence of BBD or the relative proportion of beech, but this does support the idea that timber stand improvement can improve the financial outlook of aftermath forests by increasing the potential for high-quality sawtimber. Leak (2006) compared three management approaches in the Bartlett Forest that include multiple single-tree selection cuttings since 1952 in an uneven-aged stand, an uneven-aged stand that received one cut between 1952 and 1953, and a 100-year-old stand that was commercially thinned in the 1950s. Their findings showed that uneven-aged management corresponds with a greater degree of merchantable beech trees and an improvement in the grade of sawlogs, although no costs are provided for any of the management strategies. Bohn and Nyland (2003) concluded that the development of understory beech following cutting treatments was determined by the pre-cut levels of beech, and unless a stand has previously low levels

of beech, that it is likely to become the most abundant postharvest species. Farrar and Ostrofsky (2006) reported that beech regeneration increases following winter harvests for up to 10 years after cutting.

Kochenderfer et al. (2004) reported application costs and efficacy of controlling beech with herbicides in West Virginia. The cost of herbicide application, including chemical and labor, ranged from \$97/ha for injecting 393 stems/ha greater than 15 cm in diameter, up to \$569/ha for spraying 9,249 stems/ha less than 2.5 cm in diameter. Both injection and spraying were up to 99% effective in killing beech a year after treatment. Although this method is highly effective, this can be a tremendous cost for both industrial and nonindustrial private landowners who may not have sources of revenue to offset these investments. The bottom line is that there are many factors to consider when implementing BBD remediation, and even the most effective strategies will not be implemented due to financial barriers.

This study sought to identify the circumstances that would allow a logging company to conduct a large-scale remediation harvest and generate enough revenue to cover all costs involved under a stumpage contract. Operational margins can be very narrow for many logging companies conducting regular harvests due to such factors as harvested volume, BMP implementation, stumpage contract rates, rising fuel costs, and financing terms for equipment. The economic viability of a logging job is dependent on the logging company's ability to cover fixed costs (e.g., payments for equipment, insurance) and variable costs (wages, fuel, etc.) while generating enough profit to reinvest in the company to offset the depreciation of machine value (Regula et al. 2018, Germain et al. 2019). In this study, we analyze operational characteristics that affect profit margin and return on investment (ROI), which controls the ability of a company to continue operating in the long-term. ROI is defined as the ratio of total profit to the total value lost in machine equity through operation on a particular logging job, or simply the ratio of total profit to total depreciation of all assets associated with a specific logging job. A range of market and site conditions were used to illustrate how the economic viability of a BBD remediation harvest may vary. Our specific objectives for this case study were to:

- determine the sources of cost from this method of BBD remediation.
- calculate the cost to cut, skid, and land sawtimber and roundwood to produce clean chips.
- examine the market conditions that would allow for a profitable implementation of BBD remediation.
- examine how the total cost of harvesting may change with the relative stand density of beech.

Methods

Study Area

The harvest for this case study was conducted during the winter of 2020 at the Huntington Wildlife Forest, a property owned by SUNY College of Environmental Science and Forestry within the Adirondack Park of upstate New York. The Huntington Wildlife Forest represents some of the worst aftermath forest conditions caused by BBD within its range (Morin et al. 2005). The most recent commercial operation in the selected forest units occurred in 1974, when a regeneration harvest reduced the basal area from 29 m²/ha (125 ft²/

ac) to 10 m²/ha (43 ft²/ac). Since that time, beech stems under 1.3 cm (.5 inch) in diameter increased to nearly 9,000 stems per hectare. The harvest consisted of seven paired, control-treatment blocks; however, only two such harvesting blocks were included for this study, which have a combined area of 48.63 hectares. The sawmill was located 70.7 km from the landing and the paper mill was 101.5 km away.

The harvest was located nearly 13 km from NYS Route 28N on a forest road and restricted to the winter season to reduce impacts of soil compaction and rutting. Taken together, these characteristics contributed job-specific costs to the operation through winter road preparation and maintenance that would otherwise not be realized for roadside or summer harvests. This harvest was completed using a stumpage contract in which the logger assumed the financial risk. Using a timber sale prospectus provided by the landowner, the logger submitted a stumpage bid based on unit volume that encompassed the estimated sawtimber and chipwood from the timber sale. This was not a lump sum sale, but rather a scaled sale in which periodic payments to the landowner were based on load totals from the destination mills.

Silvicultural Prescription

The prescription was effectively a modified shelterwood harvest where, in addition to the overstory reduction, all beech stems greater than 1.5 m (5 ft) tall were removed. The cutting blocks had preharvest basal areas of 39 m²/ha (170 ft²/ac), with beech comprising 11 m²/ha (49 ft²/ac). Both cutting blocks had a target basal area of 9 m²/ha (40 ft²/ac), with the stipulation that all beech stems were to be removed save for marked trees showing signs of disease resistance. The logger used a fully mechanized harvesting system consisting of two tracked feller-bunchers, four grapple skidders, two log loaders, and a flail chipper. The feller-bunchers cut and bunched marked trees and used a continuously moving saw (hot-saw) to cut small diameter beech stems. Grapple skidders then hauled the bunches to the landing where the log loaders topped and delimbed trees while sorting sawtimber from chip material. All harvested beech, along with lower grade material from other species, were fed into a flail chipper, which debarked and chipped pulpwood for paper manufacturing. Sawtimber of various commercial species, predominantly sugar maple, yellow birch (*Betula alleghaniensis* Britt.), and white ash (*Fraxinus americana* L.), were marketed to a local sawmill. Processes conducted at the landing included topping, delimbing, slashing, and chipping. Conversion factors by Lutz (2019) were used to convert sawtimber volumes to tonnes.

Daily Production Journals

To measure the productivity and efficiency of the harvest as well as the time dedicated to processing beech, a gross time-motion study was implemented using daily production journals (Christian and Brackly 2007, Kelly and Germain 2016, Kelly et al. 2017). Although on-the-ground elemental time-motion studies are commonly used to study forest harvesting (Behjou et al. 2008, Lowell et al. 2008, Pan et al. 2008), such an approach was neither practical or economical for this study due to the harvest area, location, and system. Given the number of pieces of logging equipment across the landscape, several on-site research assistants would have been required to monitor the activities of each operator over the duration of the harvest. The use of sophisticated electronics to measure the time and output by machine and task

(Szewczyk et al. 2014, Botard et al. 2015, Contreras et al. 2017, Bilici et al. 2019) was also considered, but the logging equipment represented by this harvest system was not equipped with functioning built-in microcomputers with timing software designed for chronometry. Furthermore, we did not have the resources to equip each piece of logging equipment with on-board digital video cameras, GPS receivers, high-frequency accelerometers, and external power sources for each piece of technology. Even with the technology, we would have requested that each equipment operator note when beech was being mowed, skidded, and processed on the landing.

The daily production journals were the best alternative for this study, allowing equipment operators to quickly record necessary information to calculate productive machine hours (PMH), as well as production details specific to their machine. The journals were designed to capture data of interest, but without requiring so much detail that it would contribute a significant amount of time to delays. Fields such as start and end times, maintenance delays, weather, and terrain were included for all the machine journals. Fields specific for their machine included time spent cutting noncommercial beech stems for the feller-bunchers, number of bunches and average skid distance for skidders, and proportion of beech processed by log-loaders. Additional information, including fuel consumption, wages, maintenance costs, was collected via interviews with the company owner and the maintenance supervisor.

Machine Rates

The cost analysis for this study was conducted using Planning and Analysis for Timber Harvesting (PATH) 2.1, an Excel-based software program to assist loggers understand their operational costs (Bick 2017). Machine rates, trucking costs, and overhead were calculated, and the results were used to calculate the overall financial feasibility of the harvest. Specific data field inputs for PATH 2.1 are summarized in Table 1.

Productive hours were measured using the scheduled machine hours, which was calculated using the daily start and end times for each machine and then subtracting time lost to delays and breaks. The productive hours were then compared to the scheduled hours, and the ratio of these values, the utilization rate, was subsequently calculated and entered into PATH. Equipment acquisition costs were provided by the owner along with financing arrangements for each piece of equipment. The entire regular maintenance history for each machine was provided, and a summation of these costs was divided by the total machine hours to date to calculate the repairs and maintenance costs per productive machine hour, as required by PATH. The expected annual use of each machine was calculated by dividing the lifetime machine hours by years of ownership, and the expected hours until major repairs was estimated by considering the age of the equipment and annual use. The logger provided their cost of lubricants, which was extrapolated using the production journals to estimate the cost of lube per 1,000 hours of use. Hourly operator and fuel costs were provided by the logger and entered directly into PATH. An average daily fuel consumption was estimated for each machine by the maintenance supervisor, and total fuel use for the entire job was provided by the owner. By dividing the daily use into the total use, a precise measurement for the rate of fuel consumption was entered into PATH for each machine.

Table 1. Input fields for PATH 2.1 with their descriptions and assumptions (adapted from Regula et al. 2018).

Input	Description	Assumptions and supplemental sources
Acquisition cost	The original purchase price, including taxes and delivery fees	Includes the capital costs of any improvements made to a machine at the time of acquisition (e.g., fixing up a used piece of equipment). Used to calculate the alternative return on investment for these funds for a time period equal to the useful life of the machine and then converted into an hourly rate
Financed amount	The financed portion of the acquisition cost, if any	
Opportunity cost	The alternative rate of return for any funds used for a down payment on this piece of equipment	
Loan interest rate	The loan interest rate, if financed	Allows calculation of a periodic hourly rate, rather than a lifetime rate. Depreciation of machines is steeper at the beginning of their useful life; this information accounts for that. Significant overhauls require recalculation of a new rate as they constitute further investment in the machine.
Loan term	The loan term, if financed	
Machine ownership life	The number of hours this machine can be expected to be in service in its useful life	
Machine hours until significant overhaul or repairs	The machine hours until a significant overhaul or repairs	Allows calculation of a machine rate for a segment of the machine's life, more closely matching the actual loss of machine value that occurs. PATH 2.1 uses hourly depreciation as the measure of investment in individual harvesting jobs.
Percent of cost to depreciation	The percent of the machine's value that will be depreciated between the original acquisition cost and the next significant repair	Used in converting the fixed cost of insurance and the alternative annual return for cash invested in the equipment into hourly rates
Expected annual use	The number of hours the machine is used in a year on average	Represents a fixed (non-depreciating) amount of investment in the machine. An annual return on this investment is expected. PATH 2.1 accounts for the return on this investment that is included in individual jobs.
Residual value	This is the salvage value	Supplied by loggers
Repairs and maintenance cost per productive machine hours	Hourly repairs and maintenance costs	Supplied by loggers
Fuel cost per gallon	The fuel cost per gallon	Supplied by loggers
Fuel consumption rate	This is the fuel consumption rate (gallons/hour)	Supplied by loggers
Lube costs per thousand hours of service	This includes costs of oil changes and grease.	Supplied by loggers
Hourly operator costs	This includes the operators' hourly wage, workers' compensation, and benefits.	To calculate hourly operator cost to the employer, we doubled the operator's hourly wage to include benefits.
Ratio of machine hours to operator time	This is the percent of the operator's time that is actually spent operating the machine.	Used to calculate the cost of the labor per productive machine hour. For example, if the hourly operator cost is \$30 and the ratio of machine hours to operator's time is 80%; the cost per machine hour is \$37.50 ($\$30 \div 0.8$).
Annual insurance costs	The cost of replacement insurance on the machine	Only applies if the machine was actually insured (usually as a requirement of the financing).
Daily overhead	This is used to capture other business expenses (e.g., nonproductive labor, liability insurance, advertising, pickup truck expenses, legal and professional services, office expenses, and other items typically found in IRS Form 1040 Schedule C).	
Job-specific costs	One-time costs directly associated with an individual harvest (e.g., moving costs [one way], road improvements, BMP costs)	
Days on the job	This represents all productive days on the job.	Used to calculate the total overhead associated with the job (i.e., number of days $\times$ daily overhead).
Production	This includes product name, total production, and price.	Final production volumes were verified with foresters.

Equipment insurance rates were estimated by machine based on their lifetime hours and age.

Over the course of interviews with the logger, values were determined for each category involved with overhead costs. Health insurance rates, worker's compensation, wages, rent, and pension plans were among the largest sources of overhead. After a value was agreed upon, the owner certified the estimates with their accountant, and estimates were then verified against the literature.

The acquisition cost, financing details, and mileage for the trucks involved with transporting equipment and timber products were provided by the company's owner. Then, a generic lifetime travel distance was entered for more generalizable results. The maintenance supervisor provided the maintenance interval for the trucks and an estimate for the maintenance cost per 1,000 hours. Upon entering these details for each truck involved, a cost was generated that represents the cost of depreciation for every kilometer of truck transport. Finally, the owner provided the driver wages so the total trucking costs could be calculated. The trucking costs were calculated by taking the cost per kilometer calculated by PATH and the driver wage per trip provided by the logging owner and multiplying these values by the number of trips and number of kilometers traveled by trucks.

Cost Allocation

Breaking down the unit costs of production required joint cost allocation, because sawtimber and clean chips were produced simultaneously during felling and skidding which, by definition, makes the production of these coproducts a joint cost (Blocher et al. 2002, Wagner 2012). We used a mass allocation of time to apportion the costs of production, because the machine rate method uses time to calculate cost, and the portion of time spent on either product depends on the volume being produced, not revenue being generated, which would be used for an accounting allocation of cost. However, because most of the harvested trees were chipped, converting the volume of sawtimber to mass tonnage was deemed the most accurate approach.

Sensitivity Analysis of ROI

System financial viability from the logger's perspective was assessed by comparing their costs and revenues with their total investment in the harvest. Profit was calculated by summing the overhead investment, site specific costs, machine rates including depreciation and wages, and stumpage payments to the landowner, then subtracting that value from total revenue. ROI is a more useful metric of viability than profit alone because of its implications regarding the ability of the company to reinvest in equipment, ensuring the future status of the company. ROI measures the amount of return on an investment relative to the investment's cost and is calculated by dividing profit (total revenue-total costs) by the investment (sum of fixed hourly investment costs and depreciation).

The prices for sawtimber were sourced from a price sheet provided by the hardwood sawmill, and the unit price for clean chips (chips free of bark material graded for paper production) was confirmed by the logger. Although these prices are subject to change by year or by the relationship between the mill and the logger, they are necessary to determine the financial implications of the harvest. To reflect this range of values in the market, we examined different purchasing

prices for clean chips to simulate the viability of this harvest if it were to take place in a region void of competitive chip markets. Chip markets are also not ubiquitous throughout beech's range, so we analyzed how ROI would be affected by substituting fuel chips for clean chips. Finally, the ability of a stand to generate high-quality sawtimber is variable throughout the range of BBD, so we examined the impact on ROI for producing strictly clean chips or fuel chips.

Our estimates of profit and ROI depend in part on our calculations of machine rates using PATH. We conducted a sensitivity analysis of profit and ROI to determine how they shift when the machine rates vary by intervals of 2%, up to a 10% increase. This illustrates how other logging companies might fare in this harvest if they experienced more expensive operating costs. For this analysis, we held the costs of product delivery, job specific costs, and overhead constant to elucidate the sole impact of machine costs on harvest viability.

Results

The harvest yielded 527 metric tonnes of sawtimber (99.5 thousand board feet) and 4893.9 tonnes of clean chips which were sold to the mills at an average purchasing price of \$115.17/tonne and \$57/tonne, respectively, resulting in a total revenue of \$341,161 (Table 2). The logger was under a stumpage contract with rates averaging \$44.57/tonne for sawtimber logs and \$7.17/tonne for chips, and subsequently paid the landowner a total of \$58,497 for stumpage. The total cost of harvesting was defined as the combination of machine rate costs, trucking, overhead, and job specific costs. Daily overhead was estimated at \$2,440/day, and with a total of 31 days of harvesting, resulted in a total of \$75,640. The biggest contributions to overhead costs include worker's compensation and health insurance, which must be provided for all nine members of the harvesting crew. Although there are individual choices each logging firm can take to reduce its overhead, this is largely a function of the size of the company, specifically the number of employees and price of insurance. The machine rates and trucking rates were multiplied by the

Table 2. Volume and revenue of sawtimber and clean chips harvested during a beech bark disease remediation harvest in the winter of 2020 in the Adirondack Park, New York.

Species	Volume, tonnes	Revenue, \$
Sawtimber		
Sugar maple	350.28	44,610.00
White ash	127.88	11,509.85
Yellow birch	31.09	3294.85
Red maple	9.63	678.88
Black cherry	2.88	235.63
Paper birch	3.04	176.38
American basswood	1.62	110.13
Red spruce	0.58	41.00
Total	527	60,656.72
Clean chips		
American beech	4536.62	260,014.38
Nonbeech spp.	357.32	20,489.90
Total	4893.94	280,504.28
Combined total	5420.94	341,161.00

productive hours for each machine, and the total distance traveled by trucks, to yield costs of \$135,578 for machines and \$50,539 for trucking. Finally, job-specific costs were calculated by estimating the cost of BMP implementation, road preparation, and equipment staging that are unique to this harvest. Based on the trucking rate and machine rates for the bulldozer, grader, and excavator, the machines used for these purposes, our estimate of job-specific costs was \$14,943. These combined costs total \$276,701 and, given the generated revenue, the total profit for this harvest was \$5,965 (Table 3). Investment totaled \$79,905 (Table 4), which resulted in an ROI of 7.5%.

The sensitivity analysis of machine rates illustrates how increases in the cost to operate affect the ability of the logger to generate a profit. Our results show that as little as a 4% increase in the cost to operate machinery would reduce this harvest to a break-even operation for the logger. As one might expect, the gradual increase in operating expenses results in a direct reduction in profit margin and ROI (Table 5). Operating costs that could increase the machine rate cal-

culation include the purchasing price of the equipment, the amount of the purchasing price that was financed, the repair and maintenance costs, fuel costs, operator costs, equipment insurance, and utilization rate of equipment. By coarsely increasing all machine rates by 2%–10%, incrementally, we see how marginal increases in machine characteristics can turn an otherwise profitable harvest into an operation that only partially covers costs while losing value through the depreciation of equipment. The depicted impact on ROI is a linearly inverse relationship with rising machine rates, but note that we held the investment costs constant.

Throughput, as defined by Goldratt (1991), is the rate of revenue generated through sales. In the context of this harvest, throughput is directly controlled by the rate of productivity to cut, skid, and land whole trees that will produce sawlogs and clean chips. By this definition, activities such as daily maintenance and breaks would not contribute to throughput nor would skidding slash off the landing and distributing it on skid trails, a common BMP. The task of cutting noncommercial beech stems was predicted to be a significant source of cost because of this time invested in nonthroughput activity. Of the 292.2 PMH from the feller-bunchers, 103.6 hours were spent cutting nonthroughput material, an additional cost of \$9,943 and an additional investment of \$5,137 through depreciation. This time and investment spent cutting nonthroughput material encapsulates the additional cost of this harvesting method that differentiates it from standard shelterwood harvests. Based on the density of beech interpolated by Morin et al. (2005) using Forest Inventory and Analysis data from the USDA Forest Service, beech comprises from less than 0.2 up to 3 m²/ha throughout most of its range and can be as high as 3–7 m²/ha in the densest pockets of its range. The preharvest inventory for this harvest indicated over 11 m²/ha of beech, which corresponds with the 35% of the feller-buncher time spent on noncommercial beech. Assuming time feller-bunchers spent

Table 3. Summary of revenue, costs, and profit for a nine-person logging crew to complete a beech bark disease remediation harvest in the winter of 2020 in the Adirondack Park, New York.

Category	Value, \$
Revenue	341,161
Stumpage	(58,496)
Machine cost	(135,578)
Overhead	(75,640)
Job-specific cost	(14,943)
Trucking	(50,539)
Profit	5,965

Parenthesis indicate that a value is negative.

Table 4. Summary of cost and investment, calculated by PATH 2.1, for the duration of a beech bark disease remediation harvest completed in the winter of 2020 in the Adirondack Park, New York.

Source of investment	PMH	Machine rate, \$/PMH	Cost, \$	Investment, \$/PMH	Depreciation, \$
2014 Feller-buncher	153	95.76	14,667.67	49.92	7645.40
2012 Feller-buncher	139	96.26	13,380.13	49.18	6836.02
2017 Skidder	185	84.83	15,663.10	39.68	7325.99
2011 Skidder	152	89.00	13,565.94	43.56	6638.74
2007 Skidder	146	88.27	12,898.91	41.58	6076.07
2015 Skidder	86	94.14	8073.43	45.42	3894.93
2006 Flail chipper	171	133.71	22,901.54	66.71	11,426.18
2006 Log loader	180	86.02	15,504.07	39.63	7141.61
2004 Log loader	142	77.07	10,912.12	34.74	4917.94
2000 Bulldozer	28	93.71	2623.79	98.67	2762.67
2004 Grader	27	96.06	2593.66	68.89	1860.00
2006 Excavator	50	90.79	4539.42	48.67	2433.33
2014 Utility truck	112	71.53	8011.36	27.68	3100.16
	km			\$/km	
Trucks	42,959			0.17	7473.99
	Days			\$/day	
Non-equipment property	31			12.00	372.00
	Total	145,335.13			79,905.03

producing throughput remains constant, 188.6 hours, we varied total feller-buncher PMH to examine costs and ROI when time cutting nonthroughput beech accounted for 30% of PMH and decreasing intervals of 5% thereafter, which correspond with proportional reductions in the basal area of beech. We found a linear increase in ROI of 2% for every 1.6 m²/ha reduction in basal area of beech (Table 6).

The harvesting cost realized by the logger to cut, skid, and land was \$4,651/ha. Removing the 35% of feller-buncher time that was not spent generating revenue, the cost decreases to \$4,446/ha. This suggests that the additional cost of beech removal for this harvest was \$204/ha. This estimated cost was derived by using a mass allocation of time to apportion the costs of production. For this harvest, sawtimber comprised 9.7% of the total mass of products and clean chips accounted for 90.3%. The loader operators recorded the portion of their time spent handling beech material, which averaged 87.2%. However, they spent additional time sorting and slashing nonbeech material that yielded chips. The chipper operator provided an estimate of beech tonnage that was processed daily and noted that most nonbeech material fed into the chipper was sugar maple and yellow birch. Using the species conversion factors, we estimated that roughly 37% of the time the loaders were processing nonbeech species they were still producing chips. By taking 37% of the remaining 12.8% of the loaders' time, we concluded that 92% of the loaders' PMH were spent producing chips, and 8% spent producing sawtimber. Using the proportional mass weight of each product to allocate felling, skidding, overhead, and job specific costs, and using the loaders' time to allocate landing

process costs, we calculated the cost to cut, skid, and land was \$36.63/tonne for sawtimber and \$42/tonne for chips. Another common metric for cost analysis is the cost per unit area rather than unit of product.

Substituting sawtimber for its theoretical equivalent of 449.7 tonnes of clean chips (Miles and Smith 2009) (weight of sawtimber minus bark) would reduce the revenue by \$34,880. Reflecting this change in product with the stumpage payments and comparing this to all products being delivered to the paper mill, this change in product would result in a deficit of \$9,979 and ROI of -12.4%. Similarly, if there were no sawtimber from the harvest, yet the paper mill was located next to the sawmill and the market prices for chips were consistent, the reduction in revenue would equate to \$14,668, yet the logger would still have made a profit over \$23,700 and an ROI of 30%. These results indicate that the presence of sawtimber make a BBD remediation harvest considerably more feasible, yet the proximity and availability of nearby markets is still a more influential factor based on the delivered prices.

Producing clean chips is relatively uncommon in the Northeast because few loggers own flail chippers due to their repair and maintenance costs coupled with the financial uncertainty for loggers to branch into other product systems. A much more common product for low-grade forest products is biomass or fuel chips used for energy feedstock. Delivered prices for biomass chips are markedly lower than clean chips because of their relative utility; biomass chips will soon be incinerated whereas clean chips enter the supply chain. However, the markets for both products are subject to fluctuation based on natural gas prices and demand for paper, among other factors. We analyzed the feasibility of this harvest system across product markets and market conditions by substituting the delivered price of clean chips for biomass chips, then adjusting the price by increments of \$5.51/tonne to examine the resulting profit and ROI. For this analysis, we assumed that the energy plant was located next to the paper mill and the sawmill was in its true location, and all costs were held constant with their true values. Starting with a delivered price baseline for biomass chips that is half of the commanded price for clean chips, \$28.66/tonne, the company would have generated a deficit of \$134,291 and ROI of -168%. The results of this evaluation clearly demonstrate that attempting to implement this harvest to generate biomass chips would be impractical for most land-owners unless feedstock prices inflated dramatically (Table 7). Alternatively, because the required price to generate a profit

Table 5. Incremental increases in machine rate and the corresponding changes in operating expenses, profit, and ROI for a nine-person logging crew during a beech bark disease remediation harvest in the winter of 2020 in the Adirondack Park, New York.

Machine rate increase, %	Operating expenses, \$	Profit, \$	ROI, %
0	276,701	5,965	7.5
2	279,607	3,059	3.8
4	282,514	152	0.2
6	285,421	(2,755)	-3.4
8	288,328	(5,662)	-7.1
10	291,234	(8,568)	-10.7

Parenthesis indicate that a value is negative.

Table 6. Predicted financial values for a range of American beech stocking densities assuming a linear relationship between beech stocking and nonthroughput time invested by feller-bunchers (FB) based on reported hours and stocking levels during a beech bark disease remediation harvest in the Adirondack Park, New York.

Nonthroughput time, %	FB 1 hours	FB 2 hours	Beech relative stand density, m ² /ha	Cost reduction, \$	New profit, \$	Investment, \$	ROI
35	153	139	11.4	—	5965	79,905	7.5
30	145	131	9.8	1,528	7493	79,116	9.5
25	138	124	8.2	2,930	8895	78,393	11.3
20	131	116	6.6	4,334	10,299	77,669	13.3
15	123	109	5.0	5,735	11,701	76,945	15.2
10	116	102	3.4	7,137	13,102	76,222	17.2
5	109	95	1.8	8,541	14,506	75,497	19.2
0	101	87	0.2	9,943	15,908	74,774	21.3

Table 7. Profit and ROI for a beech bark disease remediation harvest in the winter of 2020 in the Adirondack Park, NY when substituting clean chip prices for various biomass feedstock prices.

Price, \$/tonne	Profit, \$	ROI, %
28.66	(134,291)	-168%
34.17	(107,319)	-134%
39.68	(80,347)	-101%
45.19	(53,375)	-67%
50.71	(26,403)	-33%
55.12	(4826)	-6%
56.22	567	1%

Parenthesis indicate that a value is negative.

with biomass chips is nearly the price for clean chips, these results could be interpreted as the forecasted profit and ROI if clean chip prices were to plummet; even a reduction in unit price of 4% for clean chips, \$57.32 per tonne down to \$55.12, would have prevented the logger from netting a profit.

Discussion

The successful completion of this beech remediation harvest was possible due to the mechanized harvest system, local markets for clean chips, presence of some quality sawtimber, and relatively low cost of the sawtimber stumpage. The landowner's willingness to accept lower stumpage prices across most species, particularly sugar maple, provided an opportunity for an economically viable logging job that yielded a respectable profit and return on investment for the logger. Had the logger paid the median stumpage prices (NYS Division of Lands and Forests 2020), it would have resulted in a significant loss. Sawtimber stumpage price reductions allowed the logger to cover the extraneous job-specific costs associated with opening and maintaining 13 kilometers of forest roads as well as absorbing the cost of the nonthroughput activity of removing noncommercial beech. Although the landowner conceded roughly \$10,000 in stumpage revenue, they were still able to earn nearly \$60,000 from the harvest and are left with a visually pleasing residual stand of primarily sugar maple seed trees without beech dominating the understory.

The ecological efficacy of the harvest cannot be confirmed or refuted at this point, as the results shall unfold as the stand regenerates. It is important to note that this harvest occurred in the winter to reduce soil compaction, rutting, and sediment runoff to nearby streams and lakes, but it is recognized that cutting beech trees during the growing season is more effective in staving off root sprouts. Harvesting in the summer would not only decrease the number of root and stump sprouts (Farrar and Ostrofsky 2006), but warmer weather eliminates the need to idle heavy equipment that is required for winter harvesting. Less idling time results in lower costs because fewer machine hours are necessary for the same level of productivity. Regardless of the harvest season, herbicide applications may be a worthwhile addition to the prescription to ensure the understory does not revert to preharvest conditions. Consequently, the landowner may need to reinvest an estimated \$10,000 of that \$60,000 of stumpage revenue to address any lingering beech issues. The actual cost of a postharvest herbicide treatment will depend on the density of beech stems, but it will likely be on the low end (\$200/ha) of costs reported by Kochenderfer et al. (2004).

As noted, the presence of sawtimber stumpage played a key role in the economic viability of this beech remediation treatment. However, timberland in this region is often poorly stocked due to past harvesting practices and low site index, resulting in very little sawtimber. If this is the case and a harvest produces primarily chipwood, the economic viability of the operation will be dependent on the harvest system, market prices, and mill location. If the logger can produce clean chips for a pulp mill and transportation costs are reasonable, it can be worthwhile for the logger. In this scenario, the landowner would generate about \$35,000 in stumpage revenue—considerably less than \$60,000, but at least the treatment is possible, and there is enough revenue to implement potential postharvest herbicide treatments. If, however, the logger is not equipped to produce a clean chip in the woods and is limited to generating fuel chips, there is no possibility to implement the job at current chip prices. The logger would not even be able to cover fixed and variable logging costs. Even if the landowner was to forego stumpage revenues, the job would not be economically viable for the logger. And finally, if the timber sale consisted exclusively of hardwood pulpwood (estimated at 4,200 tonnes), with current delivered pulpwood prices similar to fuel chips, it would not be economically viable to conduct beech remediation. In the case of fuel chips and pulpwood, the revenues are not enough to cover logging costs. Not only will the landowner have to forego stumpage revenues but also they would have to pay the logger roughly \$200/ha and consider it a long-term forest management investment.

Conclusions

Beech is a critical component of the most common forest cover type (maple-beech-birch) in Northeast. Unfortunately, the majority of these forests are affected by BBD, resulting in deleterious effects on forest values over the long-term, including timber production, biodiversity, wildlife habitat, and recreation. Silvicultural interventions are needed to remediate these forests so they can once again provide a wide array of forest values. The high costs of remediation are discouraging many forest landowners from proactively addressing recalcitrant beech in their forests. Our results indicate that landowners and loggers can successfully work together to conduct beech remediation under the right conditions. Based on our case study, we strongly recommend a mechanized harvest system that provides the economies of scale to efficiently remove the beech from the site. For the logger to dedicate a substantial amount of time towards an activity that is not generating revenue, the landowner must be willing to sacrifice some stumpage revenue. The amount of that sacrifice will depend on the products generated by the timber sale. Higher value products, such as sawtimber and clean chips, can generate more revenue, which will allow the operation to absorb the costs of beech remediation. If those higher value products cannot be generated from the timber sale, revenues are not likely to cover the costs of the operation, although this could vary by region and market conditions. In such cases, the landowner will need to pay outright and justify the silvicultural intervention as a long-term investment.

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